

# **AIAA 2290: Ethics, Privacy and Security in AI**

## **Foundations of Data Privacy**

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**Xuming HU**

**xuminghu@hkust-gz.edu.cn**

**The Hong Kong University of Science and Technology (Guangzhou)**

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**1 Introduction to Data Privacy**

**2 Data Privacy Principles**

**3 Data Privacy Technologies**

**4 Data Privacy in Practice**

**5 Data Privacy Policies and Notices**

**6 Future of Data Privacy**



01

# Introduction to Data Privacy



# **What is Data Privacy?**

# **Why Privacy Matters?**



## What is Data Privacy?

Data privacy refers to the protection of individuals' personal information from **unauthorized access, use, or disclosure.**





## Why Privacy Matters

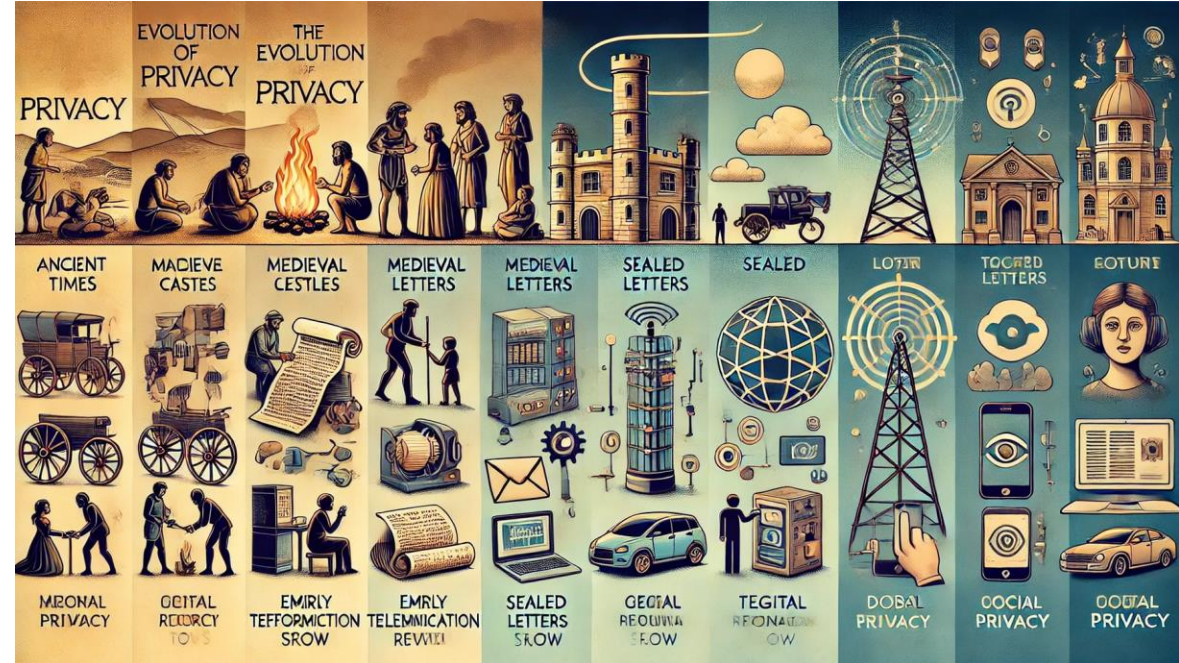
Privacy is a fundamental human right and a key component of personal autonomy. It matters because it protects individuals from potential harm, such as **identity theft**, **financial loss**, and **reputational damage**.





## The Evolution of Privacy

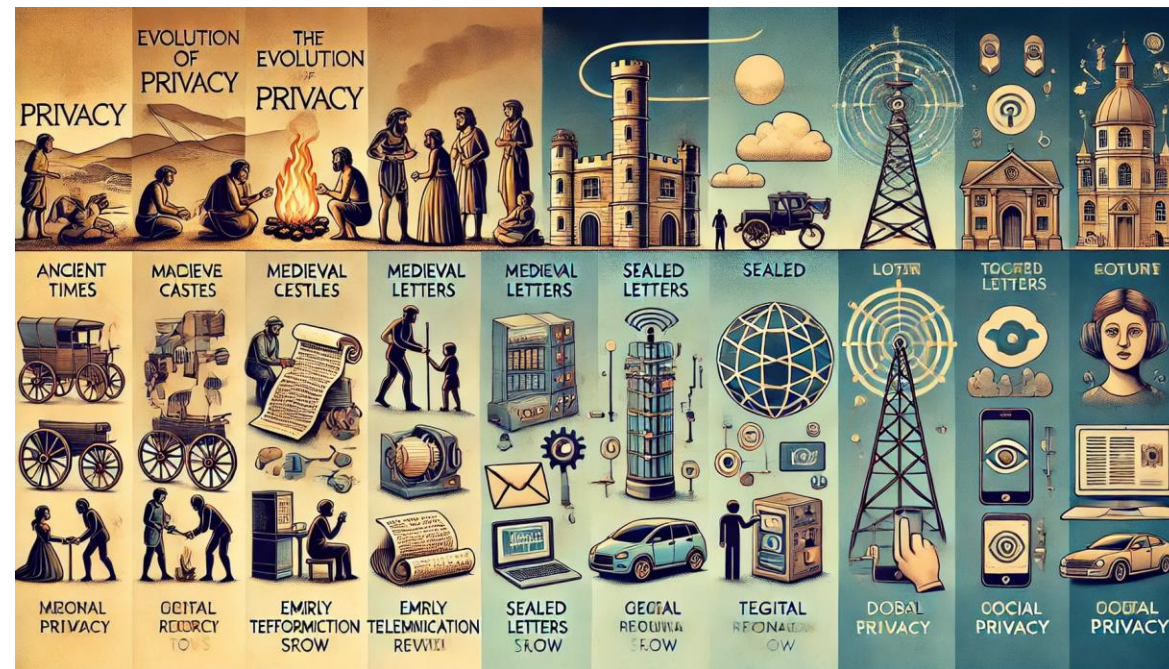
The concept of privacy has evolved significantly over time.





# Privacy in the Digital Age

**The rise of social media, e-commerce, and smart devices has increased the vulnerability of personal information.**





## *AI knowledge Quiz*

*In the digital age, the concept of privacy has become increasingly important.*

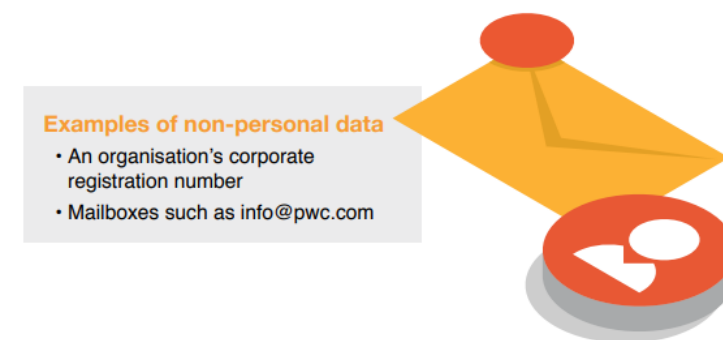
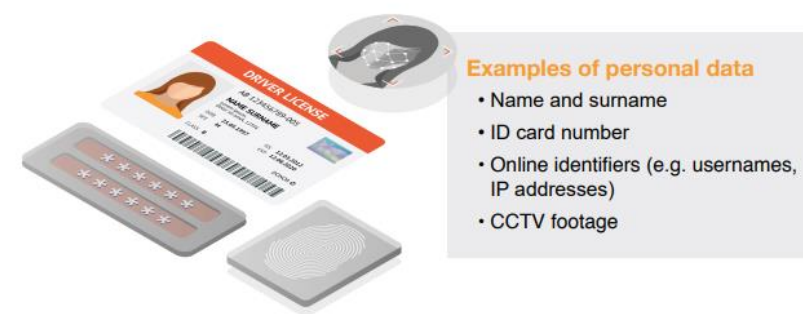
*What do you think constitutes Personal Data? Explain your answer with examples.*

*What is the difference between personal data and sensitive data?*



**Personal Data** refers to any information that can identify or be associated with a specific individual. According to various regulations (such as the **General Data Protection Regulation (GDPR)**), personal data typically includes the following categories:

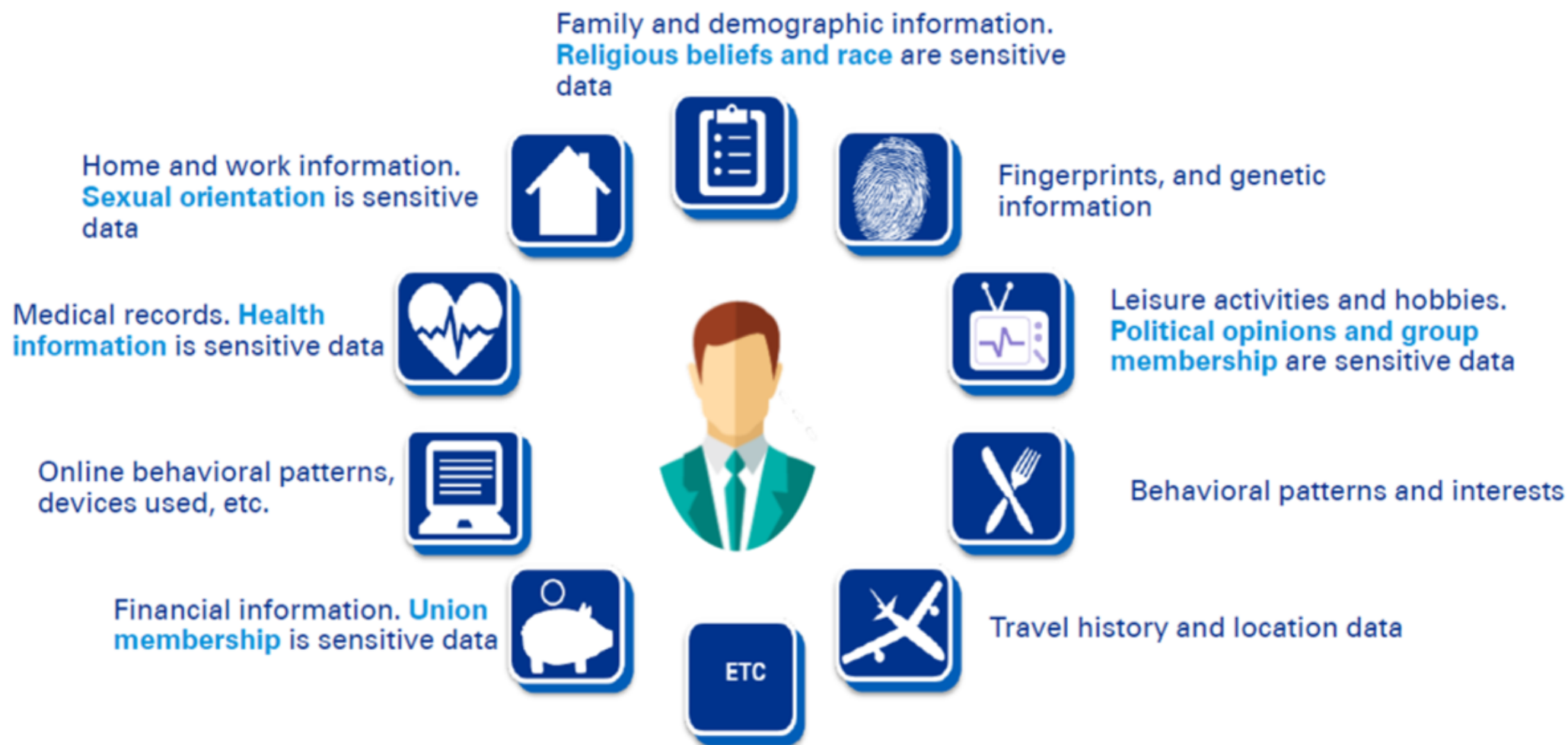
- **Directly Identifiable Information**
- **Indirectly Identifiable Information**
- **Sensitive Information**





# Some examples of Sensitive Data

**Sensitive data is a subset of personal data.**





## *AI knowledge Quiz*

*Which of the following is **NOT** considered Sensitive Data?*

*A) Health records*

*B) Race*

*C) Name*

*D) Sexual orientation*



## *AI knowledge Quiz*

*If a person's health information is disclosed, it could severely affect their privacy or security. What type of data is this?*

*A) Personal Data*

*B) Sensitive Data*

*C) Not Data*

*D) Public Data*



02

# Data Privacy Principles



**Data privacy principles** ensure that the collection, storage, use, and sharing of personal data comply with laws and regulations, protecting individuals' privacy. Below are **some key data privacy principles**:

- **Lawfulness, Fairness, and Transparency**
- **Purpose Limitation**
- **Data Minimization**
- **Accuracy**
- **Storage Limitation**
- **Integrity and Confidentiality**
- **Accountability**





## The principles are:

### **Lawfulness, fairness and transparency**

You should always process personal data in a fair, lawful and transparent manner.

### **Purpose limitation**

You should only process personal data for a specified and lawful purpose.

### **Data minimisation**

You must ensure you are only processing the personal data that you truly need and nothing more.



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### Data minimisation

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### Accuracy

You should ensure personal data is kept up to date, and that necessary measures are in place for correcting and updating inaccurate data.

### Storage limitation

You must not keep personal data for longer than you need it.

### Integrity and confidentiality

You must implement adequate security controls to ensure that personal data is protected against loss, destruction or damage.

### Accountability

You must have appropriate measures and records in place to be able to demonstrate your compliance.



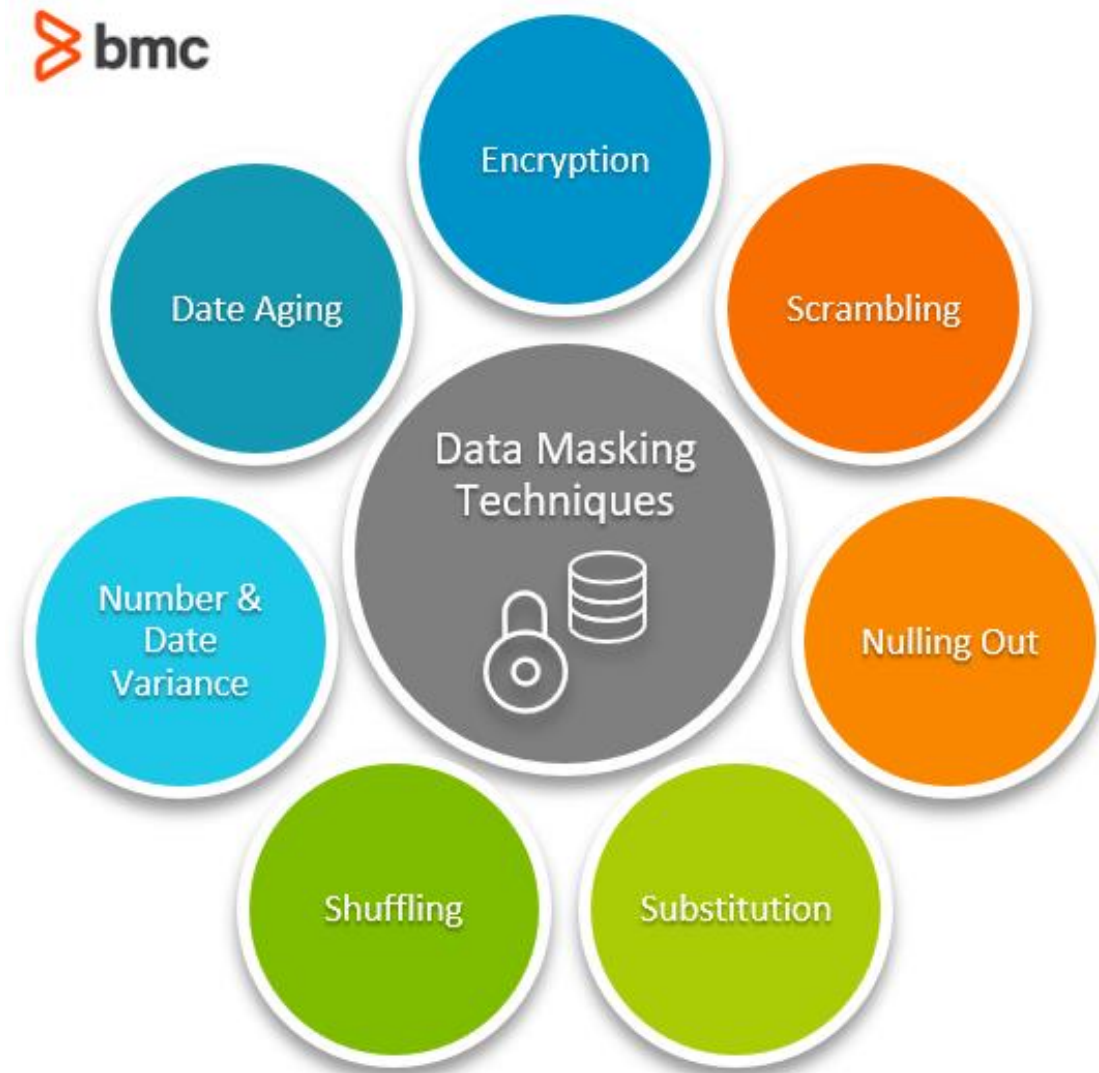
03

# Data Privacy Technologies



## Data Masking Techniques

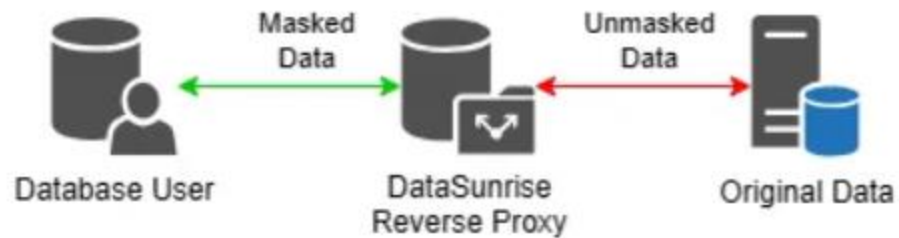
The process of hiding critical information in databases by replacing it with fictional characters or data.



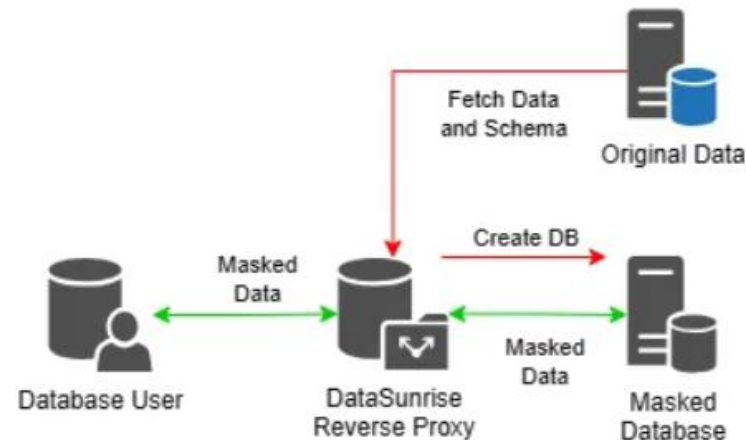


# Anonymization and Pseudonymization

- **Dynamic Data Masking** happens on-the-fly, during data retrieval.



- **Static Data Masking** masks the data before moving it to a non-production environment.



## Production Database

### Personal Informations

Patient No. 112233  
Name Peter Watson  
Address 32 Elm St  
City, State, Zip Sunnyvale, CA, 94089

### Other Info

Credit Card No. 4415 1230 0000 0062  
SSN 654 59 9876

Shuffling  
Substitution  
Custom Algorithm

Masking  
Encryption / Decryption

## Test Database

### Personal Informations

Patient No. 010101  
Name John Mayer  
Address 12 Murray St  
City, State, Zip Boston, MA, 02115

### Other Info

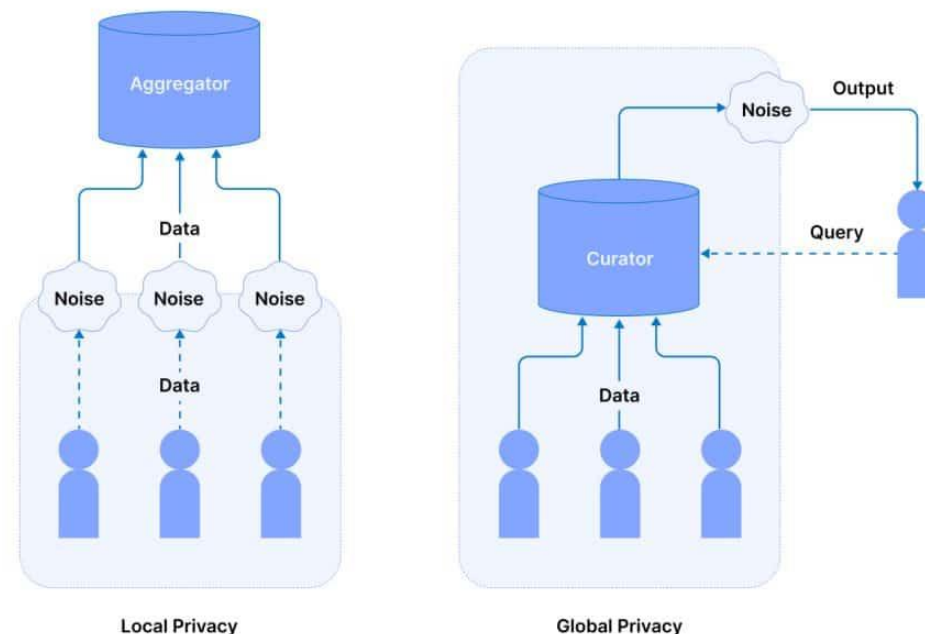
Credit Card No. XXXX XXXX XXXX 0062  
SSN @^\$%!##&#\$



## Differential Privacy

A system for publicly sharing information about a dataset by describing the patterns of groups within the dataset while withholding information about individuals in the dataset.

### How to Implement Differential Privacy





Differential Privacy - Simply Explained - YouTube





## Differential Privacy

Differential Privacy is a privacy-preserving technique aimed at ensuring that individual privacy is not exposed during data analysis or querying by adding noise to the data. Specifically, differential privacy ensures that any change to a single record in the dataset does not significantly impact the results of analysis, thus protecting the privacy of each individual. An algorithm has differential privacy if, for any two adjacent datasets (i.e., datasets differing by only one record), the output distribution of the algorithm does not change significantly. In other words, the query results from the algorithm do not allow determining whether a particular record exists in the dataset.

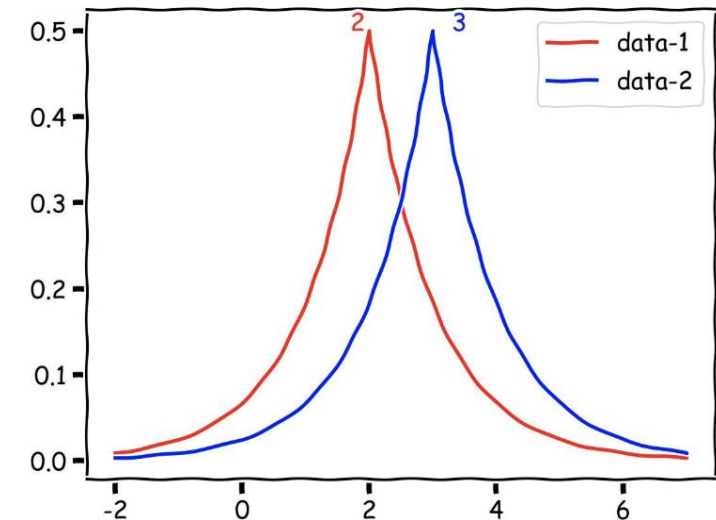
The core idea of differential privacy is to "blur" the data by adding noise (**such as Laplace or Gaussian noise**) to ensure that even if an attacker knows other data, the privacy of individual data records remains intact.



- **$\epsilon$ -Differential Privacy:** An algorithm satisfies  $\epsilon$ -differential privacy if, for any two adjacent datasets  $D$  and  $D'$ , and any possible output  $S$ , the following condition holds:

$$\Pr[\text{Alg}(D) \in S] \leq e^\epsilon \cdot \Pr[\text{Alg}(D') \in S]$$

Here,  $\epsilon$  (epsilon) is the privacy budget, indicating the level of privacy protection for the query results. The smaller the value of  $\epsilon$ , the stronger the privacy protection.





## Tokenization

Tokenization is the process of replacing sensitive data with **non- sensitive equivalent identifiers**, known as tokens, which can be used in place of the original data without exposing it

## Tokenization

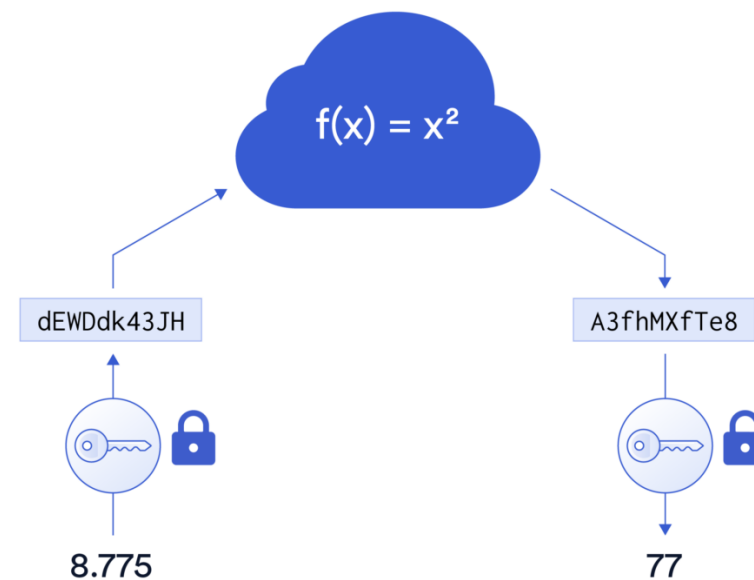




## Homomorphic Encryption

Homomorphic encryption (同态加密) is a form of encryption that allows for **computations to be carried out on ciphertext**, thus producing an encrypted result that, when decrypted, matches the result of operations performed on the plaintext.

### Compute Encrypted Data With Homomorphic Encryption





## Homomorphic Encryption

### Example:

Suppose you have two numbers,  $a = 5$  and  $b = 3$ , and you want to compute their sum without directly exposing these numbers. You can use homomorphic encryption to encrypt them and then perform the addition on the encrypted data.

1. Encrypt  $a$  and  $b$ :

$$E(a) = \text{Encrypt}(a), \quad E(b) = \text{Encrypt}(b)$$

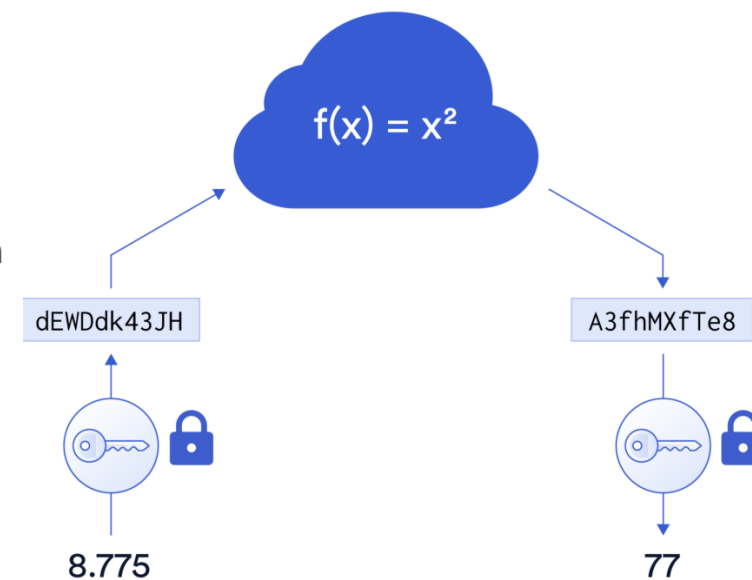
2. Then compute the sum of the encrypted values:

$$E(a + b) = \text{Encrypt}(a) + \text{Encrypt}(b)$$

3. Finally, decrypt the result to obtain the original computed result:

$$D(E(a + b)) = a + b$$

### Compute Encrypted Data With Homomorphic Encryption





## 01.

### Privacy Management Software.

Privacy management software is designed to help organizations comply with privacy regulations by automating the processes of data discovery, classification, and protection.



01.

## Privacy Management Software.

02.

## Data Loss Prevention.

Data loss prevention tools are designed to prevent data breaches by identifying, monitoring, and protecting sensitive data across networks, endpoints, and storage systems.



01.

## **Privacy Management Software.**

02.

## **Data Loss Prevention.**

03.

## **Privacy Impact Assessments.**

Privacy impact assessments are systematic processes used to identify and evaluate privacy risks associated with new projects or policies.



01.

**Privacy Management Software.**

02.

**Data Loss Prevention.**

03.

**Privacy Impact Assessments.**

04.

**Secure Messaging Apps.**

Secure messaging apps provide a platform for communication that prioritizes user privacy and data security.



04

# Data Privacy Policies & Notices



**Do you know any laws related  
to data privacy protection?**



## GDPR Overview.

A comprehensive data protection regulation that applies to all organizations processing the personal data of individuals within the European Union.



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Clear consent for data processing



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## GDPR Implementation.

- Conducting data protection impact assessments
- Updating privacy policies and notices
- Training staff on data protection responsibilities
- Implementing technical and organizational measures to secure personal data.



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## GDPR Impact on Business.

Reevaluate their data processing practices and ensure compliance with the regulation



# Legal Framework





# GDPR in Numbers

190+  
Countries  
potentially  
affected by the  
Regulation

28,000  
Estimated number  
of new DPOs  
required in  
Europe

4%  
of global  
turnover  
potential fines

80+  
New  
requirements

7  
Core data  
subjects  
rights

72  
Hours given  
to report a  
data breach



**Video Website:** What are the 7 principles of GDPR?





## *AI knowledge Quiz*

***Which of the following is NOT one of the 7 principles of GDPR?***

***A. Security***

***B. Data Minimization***

***C. Data Sharing Priority***

***D. Accuracy***



## *AI knowledge Quiz*

*Which GDPR principle requires data processing to align with the purpose of collection and not exceed the necessary scope?*

*A. Lawfulness, Fairness, and Transparency*

*B. Purpose Limitation*

*C. Data Minimization*

*D. Storage Limitation*



05

# Data Privacy in Practice



# Ten steps to an effective data privacy programme



1

Appoint a Data Protection Officer

2

Maintain a personal data register



3

Notify purpose and seek consent

4

Respond when individuals ask about their personal data

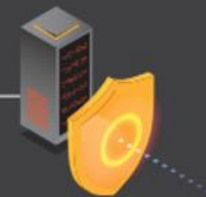


5

Enforce security mechanisms

6

Embed data privacy into your systems, processes and services



7

Notify data breaches

8

Manage third parties



9

Protect personal data when transferring overseas

10

Communicate your data protection policies, practices and processes





## Appoint a Data Protection Office

Many data privacy laws introduce the concept of a **'Data Protection Officer' (DPO)**, a new leadership role for overseeing the organisation's data protection

### Who could act as a DPO?

You can assign the role of DPO to an existing employee within your organisation, or recruit someone specifically for this role.

The DPO must be independent, an expert in data protection, adequately resourced, and must report to the highest management level.

### What's the role of a DPO?

The DPO assists you in monitoring internal compliance with the applicable data protection laws, advising you on your data protection obligations, providing expert advice when needed, and acting as a point of contact for individuals and data protection authorities.





## Maintain a personal data register

**How can I identify personal data being processed?**

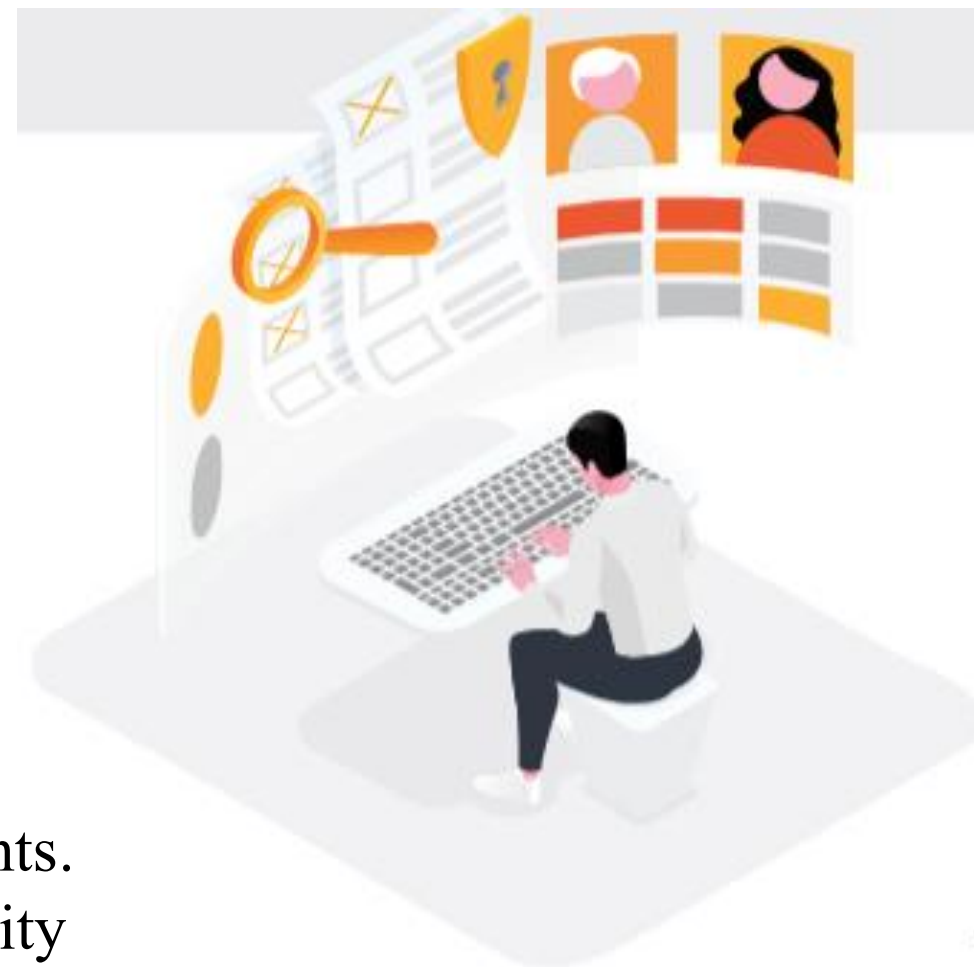




## Maintain a personal data register

### What details should I include in the register?

- Name and contact details of your DPO and any other third party (if applicable).
- The lawful basis and purpose of processing the data.
- The different categories of personal data involved.
- The systems and locations where the personal data is processed.
- Where the data is transferred to and the list of recipients.
- The retention period and enforced technical and security measures





## Notify purpose and seek consent

**Transparency is a central principle in data privacy laws.**





## Notify purpose and seek consent

### How can I obtain consent?

- Individuals can give their consent in writing or any other form. If the consent is given in writing, it should be distinct from any other agreement (e.g. terms and conditions) and written using clear and simple language.
- Individuals can withdraw their consent at any time, and the withdrawal procedure should be as easy as those for giving the consent.





## Respond when individuals ask about their personal data

- What are data subject requests?
- How can I be prepared?
- What information should I provide in my response?
- What are the steps to responding to a data subject request?

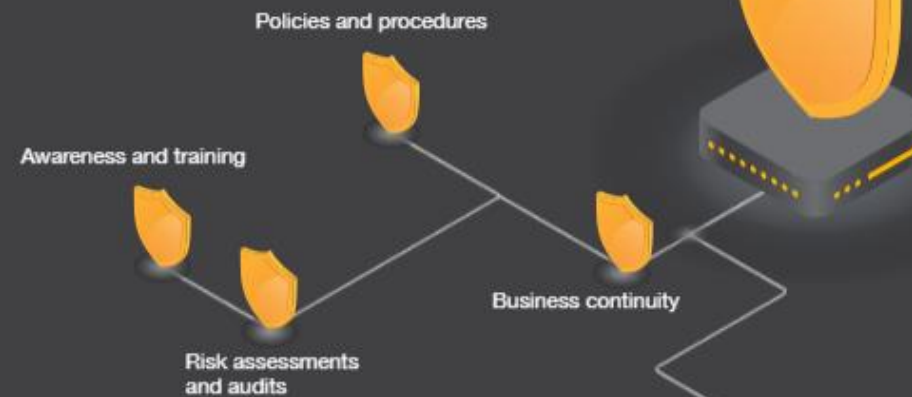




## Enforce security mechanisms

Organisations need to take reasonable steps to protect personal data.

**Organisational measures** are defined as the approach taken in assessing, developing and implementing controls that secure information and protect personal data. They can include, but are not limited to:



**Technical measures** are defined as the measures and controls implemented on systems from a technological aspect. Protecting such aspects is vital to data security, but goes above securing access to devices and systems. They can include, but are not limited to:

- System and physical security
- Encryption or de-identification of personal data
- Robust data disposal measures
- Passwords and two-factor authentication
- Bring your own device (BYOD) and remote access





## Embed data privacy into your systems, processes and services

1. Privacy and data protection are embedded into the design of a new process or application.
2. Accountability is communicated and supported.
3. Transparency is created and maintained.
4. Safeguards are established and enabled.





## Notify data breaches

### How do I respond to a data breach?

- Assess the nature of the breach and confirm if personal data is involved.
- Identify what personal data has been impacted and how.
- Assess the impact of the breach to determine if it poses high risk to the rights and freedoms of individuals.
- Determine if you need to notify the Authority and the individuals concerned.
- Carry out a thorough investigation to identify the source of the breach.





## Manage third parties

### What should I include in a contract?

- The subject-matter and duration of processing
- The nature and purpose of processing
- The type of personal data and categories of data subjects
- The minimum terms or clauses required of the processor
- The obligations and rights of the controller





## Protect personal data when transferring overseas

### What is considered a third country data transfer?

- The personal data that you intend to transfer is in scope of one or more data privacy laws.
- The personal data is transferred to a third country.
- The receiver is a separate organisation or individual. This also covers transfers to another company within the same corporate group.





## Communicate your data protection policies, practices and processes

- Compliance with data privacy laws requires that **everybody** in the organisation understands their responsibilities to protect personal data.
- It is very important to communicate your data privacy policies and practices to your **customers and employees** to ensure they are familiar with how you process and protect personal data.





06

# Future of Data Privacy



01



## IoT Privacy



01



IoT Privacy

02



Facial  
Recognition



# Emerging Privacy Issues

01



IoT Privacy

02



Facial  
Recognition

03



Deepfakes



# Emerging Privacy Issues



THE HONG KONG  
UNIVERSITY OF SCIENCE  
AND TECHNOLOGY  
(GUANGZHOU)



THE HONG KONG  
UNIVERSITY OF SCIENCE  
AND TECHNOLOGY



<https://www.youtube.com/watch?v=2k08H4MRJoQ>



# Emerging Privacy Issues

01



IoT Privacy

02



Facial  
Recognition

03



Deepfakes

04



Quantum  
Computing



### Propose New Privacy Regulations:

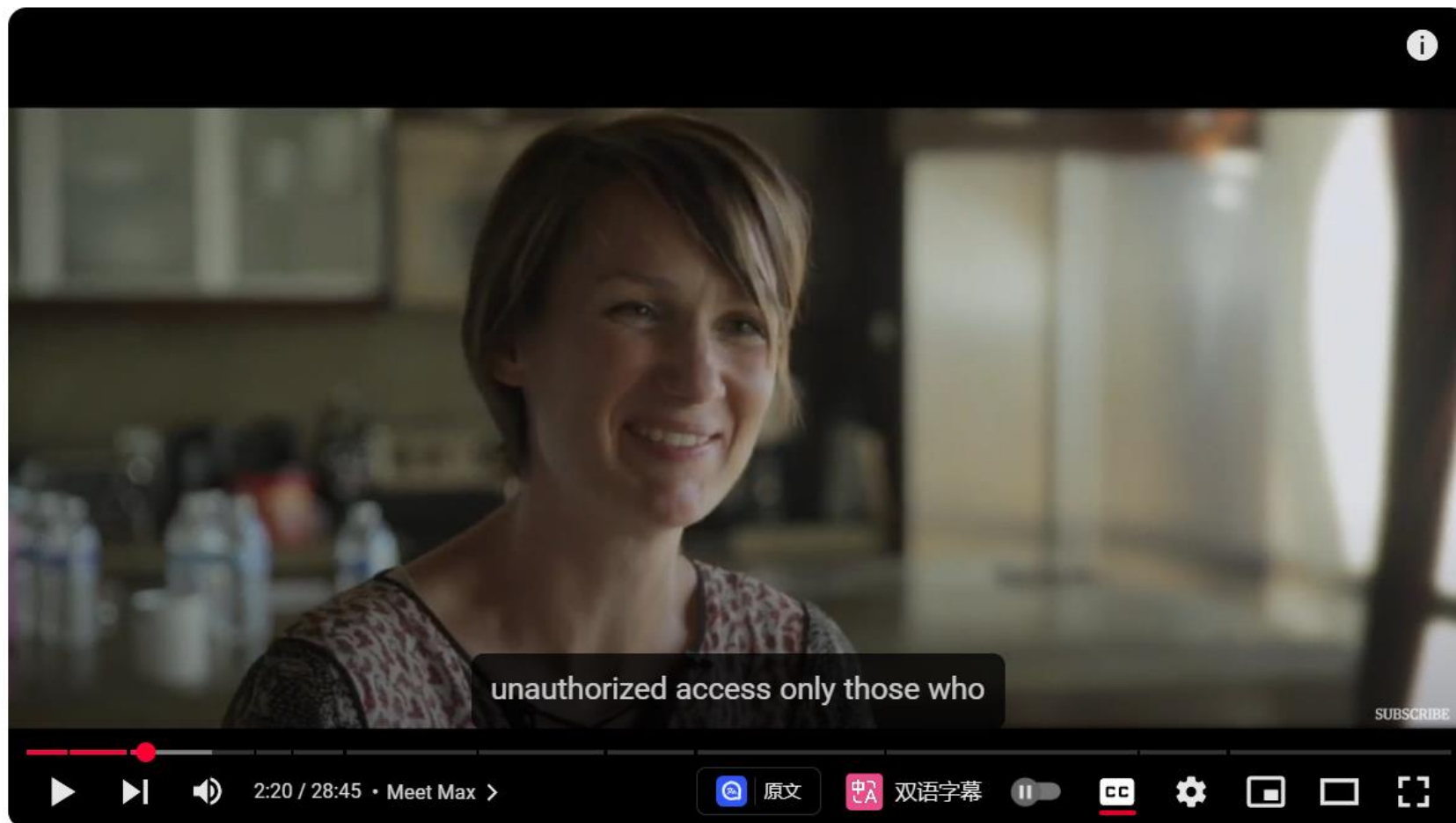
- Suggest new or modified privacy regulations to **address current data security concerns**. Regulations should aim to protect user privacy without impeding technological innovation.
- Use AI tools like **ChatGPT or Deepseek** to brainstorm ideas, analyze privacy risks, or refine your proposed regulations.
- Upload your proposal **on Canvas (10 minutes)**.
- Anyone who actively shares their AI insights with the class will earn a small bonus reward.
- **Practice is not included in the final score.**



If you are interested, you can watch this documentary after class.

In this film, Aleks Krotoski travels the world to undergo challenges that explore our digital life in the 21st century. Watch her be stalked and hacked, fight to get leaked documents back, dive into open data and live in a futuristic home that monitors her every move.

**Video Website:** <https://www.youtube.com/watch?v=KGX-c5BJNFk>



The Power of Privacy – documentary film

# **AIAA 2290: Ethics, Privacy and Security in AI**

**Thank you**

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